



The modeling of dissimilar welding of immiscible materials by using a phase field method

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ABSTRACT

A multiphysical model of high power beam welding of immiscible materials is developed to explain the influence of operational parameters and materials properties on resulting morphology by simultaneous solving of heat transfer, fluid flow and mass transfer problems. The introduction of phase field description of the interface motion between two immiscible liquids allows obtaining the cartography of melted zone in function of two key-parameters: the position of heat source relatively to joint line and the welding speed.

Due to the short thermal cycle limiting mass transfer, high power beam welding techniques may result in very inhomogeneous melted zones. In this study, the interest is paid to copper to stainless steel Nd:YAG laser and electron beam joints, which often present the repetitive heterogeneous patterns. The modeling results are validated by the comparison with experimental data on weld shape, composition and morphology.

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1. Introduction

The understanding of transport mechanism in melted zone during the welding of dissimilar metals provides an important information on mixing of the materials and consequently on the distribution of species and resulting phase content. The local composition of dissimilar weld depends on the nature of dissimilar couple (full or limited solubility, formation of intermetallic phases, etc.), on the duration of thermal cycle and on convective transport. The difference in physical properties may lead to weldability problems caused by alternation of fusion and solidification processes between the materials. The high power sources like electron beam and laser deal successfully with dissimilar joining by the application of high temperature gradients in very local volume [1,2].

The modeling of dissimilar welding is relatively new field of research. The first models proposed by Wei and Chung [3,4] and Phanikumar et al. [5] based on volume of fluid and enthalpy methods, were dedicated to laser dissimilar welding. The authors underlined the asymmetric development of the melted zone and the necessity to optimize the position of heat source in order to achieve effective mixing. Chakraborty and Chakraborty [6] presented the model of copper-nickel laser welding by Reynolds Averages Navier Stokes simulations, where the convection in the melted zone was assumed as a turbulent ($\kappa - \epsilon$ model). Dörfler [7] proposed the model of dissimilar friction stir welding where level set method was used for tracking the sharp interface.

Present study is dedicated to the modeling of high power beam dissimilar welds having coexisting immiscible zones. Insufficient mixing, which may be harmful for weld quality, is a result of short lifetime and slow convection in melted zone, of the immiscibility of components or of very sharp difference in phase change temperatures. The major specificity of high powered beam interaction with metallic materials consists in extremely high energy input in very localized area, which

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Nomenclature

$flc2hs$	smoothed Heaviside function (in COMSOL Multiphysics)
A_r	atomic mass
A_l	material property in liquid phase
A_s	material property in solid phase
c_p	heat capacity
d	offset of keyhole from joint line
dT	interval of phase change (melting)
dV_m	interval of volume fraction change at copper–steel interface
k	thermal conductivity
L	length of melted zone
$\vec{n}$	normal to the surface
P	beam power
P_l	pressure in liquid phase
R	universal gas constant
Re	Reynolds number
S_{copper}	surface of melted copper
S_{steel}	surface of melted steel
T	temperature
t	time
T_0	ambient temperature
t_{eq}	lifetime of undercooled melt
T_f	fusion temperature
T_i	solidification front temperature
t_m	melting time
t_{ini}	time of initialization of phase field function
T_{vap}	vaporization temperature
u	x-component of velocity field
U	global velocity field
U_r	velocity field relative to keyhole
U_w	tangential velocity
v	y-component of velocity field
v_0	maximal velocity of solidification front
v_c	cooling gradient
v_s	velocity of solidification front
V_m	volume fraction of material issued from phase field calculation
v_w	linear velocity of heat source
$\emptyset$	beam diameter
χ	mobility turning parameter
ΔH	latent heat of fusion
ΔL	step of heterogeneous repetitive pattern in horizontal weld morphology
ΔT	undercooling
ε	thickness of moving liquid/liquid interface
ε_m	emissivity of metallic surface
ϕ	phase field function
ϕ_0	phase field function when $t = 0$
γ	mobility of moving liquid/liquid interface
φ	atomic fraction
φ_{Cu}	calculated copper fraction in the weld
λ	mixing energy density
σ	Stefan–Boltzmann constant
$\sigma_{L1/L2}$	interfacial tension at liquid/liquid interface
η	dynamic viscosity
ρ	density

results in formation of deep and narrow cavity filled with metallic vapor (a keyhole) surrounded by relatively thin liquid layer [2]. The development of thermocapillary convection is often limited due to a weak ratio between liquid film thickness and keyhole diameter [8]. Reynolds numbers in resulting melted zones are often situated in the domain of recirculation flows and sinusoidal waves [9]. The mixing of the materials is controlled by relatively slow convection around the keyhole.

Consequently, resulting welds have well localized heterogeneous zones. Low welding speed associated with high energy input may result in formation of repetitive patterns composed by overlapping melted zones [8]. In case of dissimilar welding, it leads to the formation of repetitive heterogeneous patterns.

In the present study, the finite element based COMSOL Multiphysics software package was chosen for the implementation of our models, because of the flexibility in formulation of complex multiphysical problems by the combination of pre-defined blocks of PDE-equations, by the possibility of use of numerous stabilization techniques, by the large choice of solvers, high accuracy and comfort of use. In present, this product (originally called FEMLAB) rivals successfully with the leaders of scientific computation as, for example, ANSYS and Fluent. In 2004, Verdier [10] carried out the benchmark tests of these products and shown that the performances of FEMLAB are comparable, both from precision, CPU time and memory usage, to those of ANSYS, except from eigenfrequency analysis, and that the accuracy of FEMLAB surpasses that of Fluent (for the same number of degrees of freedom). Hanke [11] reported that, comparing to Fluent, FEMLAB needs much less degrees of freedom for a comparable accuracy. FEMLAB uses much more memory than Fluent 6.1, but the results are more accurate and the time of calculation is shorter.

In last years, many application modes dealing with moving interface and multiphase flow were added in COMSOL Multiphysics, which have made of it an attractive tool for solving the convection problems between immiscible liquids [12]. Among them are level set (first proposed by Osher and Sethian [13]) and phase field (introduced by Fix [14] and Langer [15]) methods. The comparison of level set COMSOL package with volume of fluid method used in Fluent, showed that both of them have their positive and negative sides: when Fluent provides better final solution, COMSOL allows better intermediate solutions [16]. Concerning the phase field COMSOL package, which appeared in 2009, the comparison with other software has not been done yet, however Comsol Inc. underlines its higher robustness, accuracy and easiness to couple to other types of physics, comparing to level set method [12].

Our first attempts to model the morphology of dissimilar joints with heterogeneous patterns were based on pseudo-stationary formulation of heat transfer and convection problems, associated with level set method describing the coexistence of immiscible liquids [17]. The model allowed reproducing the shape, global composition and main tendencies of heterogeneous morphology of melted zone. However, no information about the periodical structure was provided. The precision of mixing process simulation was limited as surface tension between immiscible liquid domains was not considered directly.

In present article, we propose strongly coupled multiphysical model based on temporary formulation of heat transfer, convection and mixing during the development of melted zone. We have chosen copper-stainless steel couple which has a sharp difference in thermal conductivity, melting temperature and very limited solubility associated with presence of miscibility gap in Cu-Fe system. In order to provide more physical description of mixing process, phase field method has been used for the calculation of macro-level morphology. The results of pseudo-stationary and temporary formulation of heat transfer allow obtaining the information about periodicity of heterogeneous patterns. To validate the results of the modeling, two joints carried out by Nd:YAG laser and electron beam have been used.

2. Materials and methods

The present model is created in the software package COMSOL Multiphysics 3.5. In this study, General Heat Transfer, Laminar Navier–Stokes and Phase Field tools have been used. A nonlinear direct solver UMFPACK was chosen.

The calculations were carried out with Windows XP x64, Intel(R)Core(TM)2 Quad Q8200 2.33 GHz, 8 Gb RAM.

The butt copper-AISI 316L stainless steel joints used for the validation of the model were carried out under following conditions:

- 2 mm thick plates, centered keyhole position;
- electron beam: $P = 1.125$ kW, $v_w = 0.9$ m/min, $\varnothing = 400$ μ m, vacuum chamber;
- Nd:YAG laser: $P = 2$ kW, $v_w = 0.5$ m/min, $\varnothing = 200$ μ m, argon atmosphere.

The main difference between these two techniques consists in effectiveness of interaction with metallic materials. Electron beam welding ensures almost 90% effective energy transfer [9], when Nd:YAG laser rarely exceeds 30%. Taking in consideration a weak absorption coefficient of Nd:YAG laser photons by metallic surface [2], the effective linear energy in both cases is close to 70 kJ/m, which allows attain the full penetration of 2 mm thick plates.

3. Theory and calculations

3.1. Heterogeneity of dissimilar welds

Three types of heterogeneity can be defined in the dissimilar weld:

- at the micro-level (microstructure), due to the local fluctuations of composition and temperature gradient;
- at the macro-level (melted zone), due to weak mixing between the materials;
- the directional heterogeneity, due to the fluctuation of composition along the joint line.

In the present article, the attention is paid to macro-level and directional heterogeneity. The heterogeneity of melted zone is controlled by the proportion between the materials, by the convection movements and by the shape of the melt. In case of linear advancement of the heat source, directional heterogeneity, often observable as a repetitive pattern, occurs if, within a certain temperature range, the supply of energy increases more rapidly than the heat losses. For liquid metallic systems, it leads to the local undercooling of the melt after removing of the heat source, when the new melting cycle starts near the edge of crystallized material. The melted zone remains liquid till the end of relaxation period:

$$t_{eq} = \frac{\Delta H}{C_p^l \cdot v_c} + \frac{\Delta T}{v_c}, \quad (1)$$

where ΔH – latent heat of fusion, C_p^l – heat capacity of liquid metal, ΔT – maximal undercooling, v_c – cooling gradient. If considering maximal undercooling of 200 K and average speed of cooling about 10^4 K/s, the lifetime of undercooled melt can be estimated to 0.02 s.

In such conditions, according to Frenkel–Wilson law, the velocity of solidification front v_s is not stationary and reaches its maximum when undercooled melt attends the equilibrium temperature of phase change [8]:

$$v_s = v_0 \cdot \left[1 - \exp \left(-\frac{\Delta H}{R \cdot T_f} \cdot \frac{T_f - T_i}{T_i} \right) \right], \quad (2)$$

where v_0 – maximal front velocity; T_f – fusion temperature of material; T_i – temperature of solidification front determined by heat losses and heat of fusion; R – ideal gas constant. The discernible repetitive patterns can be observed when the velocity of heat source is less than the velocity of solidification front ($v_w < v_s$) [8]. Under more important v_w , melted zone can be considered as pseudo-stationary, which means it does not present directional heterogeneity: the lengthwise composition is determined by species distribution in the tail of melted zone.

The step of repetitive pattern can be formulated as:

$$\Delta L = v_w \cdot t_m, \quad (3)$$

where t_m – the duration of thermal cycle before the start of the next melted zone and v_w – the linear velocity of heat source.

Thus, the mixing process between the components is composed of two stages: the development of melted zone during the time t_m and the retention of melted zone in liquid state due to undercooling during the time t_{eq} :

$$t = t_m + t_{eq} = \frac{\Delta L}{v_w} + \left(\frac{\Delta H}{C_p^l \cdot v_c} + \frac{\Delta T}{v_c} \right). \quad (4)$$

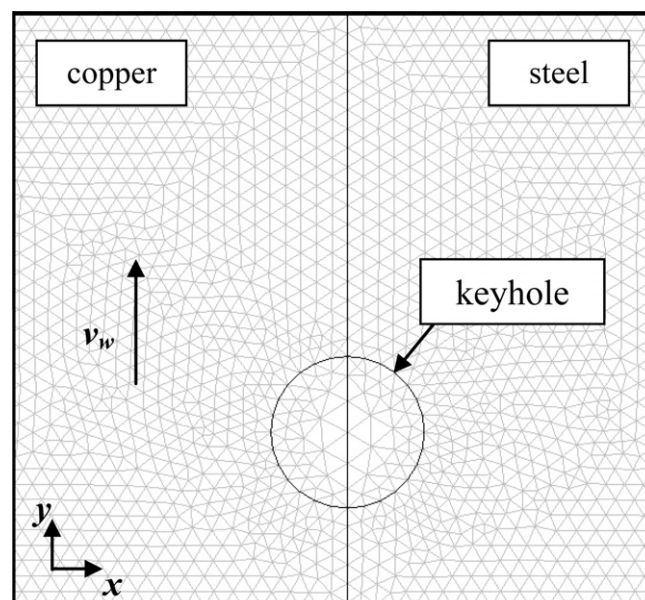


Fig. 1. Model geometry (meshing 50 μm).

3.2. Model description

The model that we propose allows reproducing the heterogeneity of a single melted zone at horizontal slice. The repetitive patterns can be reconstructed indirectly by overlapping the melted zones with a step ΔL .

Except for the top part of the joint, which may be widened, the morphology of fully penetrated high power beam welds does not change significantly with penetration as horizontal convection flows dominate. That allows reducing the geometry of the model from 3D to 2D (Fig. 1: horizontal plane) and thereby lightening the calculation. However, such reduction limits the application of the model to the cases of uniform energy distribution along z axis, when the energy supply is sufficient for full penetration, and it does not take in consideration the top part of the weld, where, generally, the vaporized zone is larger.

The modeling of such complex process like welding inevitably requires numerous simplifications. Creating the model, the heat source is approximated as a cylinder having the vaporization temperature, which horizontal projection is a circle with a diameter of the beam $\varnothing$. If heat source profile varies significantly with penetration, a careful choice of vaporization profile must be made to ensure correct results.

The role of latent heat as a second heat source is neglected: it is insignificant ($<10^2$ J) comparing to global energy input (about 10^6 J). Temporary heat transfer is assumed:

$$\rho \cdot c_p \cdot \frac{\partial T}{\partial t} + \vec{\nabla} \cdot (-k \cdot \vec{\nabla} T) = \rho \cdot c_p \cdot \vec{U} \cdot \vec{\nabla} T, \quad (5)$$

where $\vec{U} = (u, v)$ - global velocity field.

Limit conditions are:

- front surface: $T_0 = 300$ K;
- side surfaces: radiation heat loss: $\vec{n} \cdot (-k \cdot \vec{\nabla} T) = \varepsilon_m \cdot \sigma \cdot (T_0^4 - T^4)$;
- back surfaces: $-\vec{n} \cdot (-k \cdot \vec{\nabla} T) = 0$;
- keyhole profile: $T = T_{vap}^{steel}$ or T_{vap}^{copper} respectively to chosen side.

The flow past the keyhole is assumed as a single convective factor of mixing. Melted zone is considered as incompressible Newtonian liquid whose flow describes by Navier–Stokes equations:

$$\rho \cdot \frac{\partial \vec{U}}{\partial t} + \rho \cdot (\vec{U} \cdot \vec{\nabla} T) \cdot \vec{U} = \vec{\nabla} \cdot (-P_l + \eta \cdot (\vec{\nabla} \vec{U} + (\vec{\nabla} \vec{U})^T)), \quad (6)$$

$$\vec{\nabla}(\rho \cdot \vec{U}) = 0;$$

Linear displacement v_w is applied to the sides of the workpiece. The sliding condition is applied to the front sides of the keyhole $\vec{n} \cdot \vec{U} = 0$ with $\vec{n} = (n_x, n_y)$. The tangential sliding is assumed past the keyhole: $\vec{n} \cdot \vec{U} = 0$, with $\vec{U} \cdot \vec{\tau} = v_w$ at the left side and $\vec{U} \cdot \vec{\tau} = -v_w$ on the right side $\vec{\tau} = (\vec{n}_x, \vec{n}_y)$.

Resulting velocity field $\vec{U} = (u, v)$ corresponds to the superposition of linear heat source velocity of v_w and liquid velocity related to the flow around the keyhole. The last one controls the mixing between dissimilar materials and can be defined as $\vec{U}_r = (u, v - v_w)$.

To describe the movement of the interface between two immiscible liquid domains, the conservative phase field equation coupled with velocity field relative to keyhole $\vec{U}_r$ is used:

$$\frac{\partial \phi}{\partial t} + \vec{\nabla} \cdot (\phi \cdot \vec{U}_r) = \vec{\nabla} \cdot \left(\frac{\gamma \cdot \vec{\lambda}}{\varepsilon^2} \right) \cdot \vec{\nabla} \psi, \quad (7)$$

$$\psi = -\vec{\nabla} \cdot \varepsilon^2 \cdot \vec{\nabla} \phi + (\phi^2 - 1) \cdot \phi + \left(\frac{\varepsilon^2}{\lambda} \right) \cdot \frac{\partial \phi}{\partial t},$$

where ϕ is the phase field function, the mobility of the interface is defined as $\gamma = \varepsilon \times \chi^2$, ε is the thickness of the interface equal to mesh size, χ is the mobility turning parameter, $\vec{U}_r = (u, v - v_w)$ is the velocity field and the mixing energy λ is defined from:

$$\sigma_{L1/L2} = \frac{2 \cdot \sqrt{2} \cdot \lambda}{3 \cdot \varepsilon}, \quad (8)$$

where $\sigma_{L1/L2}$ is the interfacial tension between liquid domains (estimated as 0.2 N/m for Cu–Fe system [18]).

The mobility turning parameter χ can be interpreted to govern the diffusion-related timescale of the interface. It is advised [19] to keep this parameter large enough to maintain constant interface thickness but still low enough to not damp the convective motion. The physical basis of mobility turning parameter at macroscopic level can be defined as a ratio of the velocity field magnitude to the force acting at liquid/liquid interface:

$$\chi_l = \frac{|U_r|}{\sigma_{L1/L2}} \quad (9)$$

If considering the order of velocity field of 1 mm/s, resulting mobility turning parameter ($0.005 \text{ m} \times \text{s/kg}$) appears to reflect well enough the interface movement.

Phase field limit conditions are:

- $\phi_0 = 1$ in copper domain and $\phi_0 = -1$ in steel domain, which corresponds to volume fraction $V_m = 1$ and 0 respectively:
 $V_m = (1 + \phi)/2$;
- $\phi_0 = 1$ at front and side surfaces in copper domain and $\phi_0 = -1$ in steel domain;
- the contact angle between liquid metal and keyhole is assumed to $\pi/2$;
- the condition for the moving interface is given as:

$$\vec{n} \cdot (\vec{N}_{copper} - \vec{N}_{steel}) = 0, \quad \vec{N} = \varepsilon \cdot \vec{\nabla} \phi - \phi \cdot (1 - \phi) \cdot \frac{\vec{\nabla} \phi}{|\vec{\nabla} \phi|} - \phi \cdot \vec{U}_r. \quad (10)$$

To allow realistic reproduction of mixing process, it is needed to program the change of materials properties in function of local composition and phase state. For this purpose the smoothed Heaviside functions (*flc2hs*) of temperature (phase change) and volume fraction (mixing) were used. The evolution of materials properties within each Heaviside function is given by two regions, where constant value of each variable is conserved, separated by smoothing interval in order to avoid the discontinuity. For example, a function of temperature can be given as: $A = A_1 + (A_2 - A_1) \bullet flc2hs(T - T_f, dT)$. For describing the variables as density ρ , thermal conductivity k , heat capacity C_p , and viscosity η following scheme is applied:

- separation of liquid and solid domains (limited by T_f^{steel} at steel side and T_f^{copper} at copper side):

$$A_{steel} = \begin{cases} A_s^{steel} & \text{if } T < T_f^{steel} \\ A_l & \text{if } T > T_f^{steel} \end{cases}; \quad A_{copper} = \begin{cases} A_s^{copper} & \text{if } T < T_f^{copper} \\ A_{lm} & \text{if } T > T_f^{copper} \end{cases}; \quad (11)$$

- variation of liquid domain properties in function of local volume fraction:

$$A_l = \begin{cases} A_l^{steel} & \text{if } V_m < 0.5 \\ A_l^{copper} & \text{if } V_m > 0.5 \end{cases}. \quad (12)$$

As result, the evolution of melted zone shape can be located after the position of fusion isotherms of corresponding materials. The position of copper–steel interface corresponds to $\phi = 0$.

The mobility turning parameter χ is also introduced as a Heaviside function of temperature: the value issued from Eq. (9) is considered for liquid copper/steel interface; to ensure numerical stabilization a small value is applied to the solid domain:

$$\chi = \begin{cases} \chi_s & \text{if } T < T_f^{steel} \\ \chi_l & \text{if } T > T_f^{steel} \end{cases}. \quad (13)$$

Table 1
Parameters used in calculations.

Parameter	Unit	Name	Materials	
			Copper	AISI 316 L
Fusion temperature	K	T_f	1356	1720
Vaporization temperature	K	T_{vap}	2835	3013
Density (solid)	kg/m ³	ρ_s	8700	7980
Density (liquid)	kg/m ³	ρ_l	7940	7551
Heat capacity (solid)	J/(kg K)	C_p^s	385	433
Heat capacity (liquid)	J/(kg K)	C_p^l	350	734
Thermal conductivity (solid)	W/(m K)	k_s	400	8.116
Thermal conductivity (liquid)	W/(m K)	k_l	140	12.29
Dynamic viscosity (liquid)	Pa s	η_l	0.003	0.005
Latent heat of fusion	J/kg	ΔH	13.26×10^3	13.81×10^3
Interval of phase change during the melting	K	dT		50
Interval of phase change across the moving interface	–	dV_m		0.1
Interfacial surface tension between immiscible phases Cu and γ -Fe	N/m	$\sigma_{L1/L2}$	0.2	
Thickness of moving interface	m	ε	5×10^{-5}	
Mobility turning parameter	m $\times$ s/kg	χ	$\chi^l = 0.005$	
			$\chi^s = 1 \times 10^{-4}$	
Emissivity	–	ε_m	0.5	0.4
Stefan–Boltzmann constant	W/(m ² K ⁴)	σ	5.67×10^{-8}	
Beam diameter	m	$\varnothing$	2×10^{-4} (laser)	
			4×10^{-4} (electron beam)	

The parameters used in calculations are given in Table 1. Such simplified representation of materials properties is determined, above all, by the insufficient and divergent literature data on evolution of physical properties in melted metals and alloys.

For solving the model, a correct choice of calculation time is important. While t_{eq} can be estimated from physical properties of materials, t_m is unknown. Firstly, it can be obtained from measure of ΔL (Eq. (3)). Another way is to found the time step when the thickness of calculated melted zone attains the experimental value, which signifies $t = t_m$. But these options exclude predictive application of the model. If no experimental data are accessible, early pseudo-stationary formulation of heat transfer [17]:

$$\begin{aligned} \rho \cdot C_p \cdot \vec{U} \cdot \vec{\nabla} T + \vec{\nabla} \cdot (-k \cdot \vec{\nabla} T) &= 0, \\ \rho \cdot (\vec{U} \cdot \vec{\nabla} T) \cdot \vec{U} &= \vec{\nabla} \cdot (-P_I + \eta \cdot (\vec{\nabla} \vec{U} + (\vec{\nabla} \vec{U})^T)), \\ \vec{\nabla} \cdot (\rho \cdot \vec{U}) &= 0, \end{aligned} \quad (14)$$

may provide the information about the final thickness of the weld, as the calculated shape of the melted zone were shown to be in good correspondence with experiment [17]. In present study, a length measured between joint line and fusion isotherm at steel side is chosen as a reference point, which allows determining t_m .

The solving scheme consists of such steps (Fig. 2):

- (1) the calculation of pseudo-stationary heat transfer problem (Eq. (14)) [17] with stationary nonlinear UMFPACK solver;
- (2) the calculation of transient problem:
 - (a) the initialization of phase field function over a time $t_{ini} = 5 \times t$, which must be big enough to provide good smoothing of ϕ -function over the liquid/liquid interface;
 - (b) the coupled solving of heat transfer, Navier–Stokes and phase field problems (Eqs. (5)–(7)) with transient nonlinear UMFPACK solver till the time step when the isotherm of fusion at steel side fits pseudo-stationary solution, which corresponds to the end of melting (time t_m);
 - (c) the coupled solving of Navier–Stokes and phase field equations with transient nonlinear UMFPACK solver over time t_{eq} (basing on solution obtained in the last time step t_m).
- (3) the post-processing:
 - (a) temperature, velocity and phase field data are provided by COMSOL post-processing tools;
 - (b) Knowing t_m , it is possible to calculate the step of repetitive pattern (Eq. (3)) and so to reconstitute the entire morphology of a horizontal cut;
 - (c) the evolution of melted area corresponding to each material allows to determine copper volume fraction in melted zone as:

$$\varphi^{Cu}(\text{at } \%) = \frac{S_{copper} \cdot \rho_s^{copper}}{A_r^{Cu} \cdot \left(\frac{S_{copper} \cdot \rho_s^{copper}}{A_r^{Cu}} + \frac{S_{steel} \cdot \rho_s^{steel}}{A_r^{steel}} \right)} \cdot 100. \quad (15)$$

where S - area of melted material; A_r - atomic weight of copper or average atomic weight for steel, calculated as $A_r^{steel} = \varphi^{Fe} \cdot A_r^{Fe} + \varphi^{Cr} \cdot A_r^{Cr} + \varphi^{Ni} \cdot A_r^{Ni}$.

Present model reproduces the evolution of melted zone basing on relatively limited data on materials properties, which gives it an advantage comparing to enthalpy methods. Phase field method allows both to easily locate the interface between coexisting immiscible zones and to reproduce the zones of intense mixing, basing on fixed Cartesian grid. The precision of

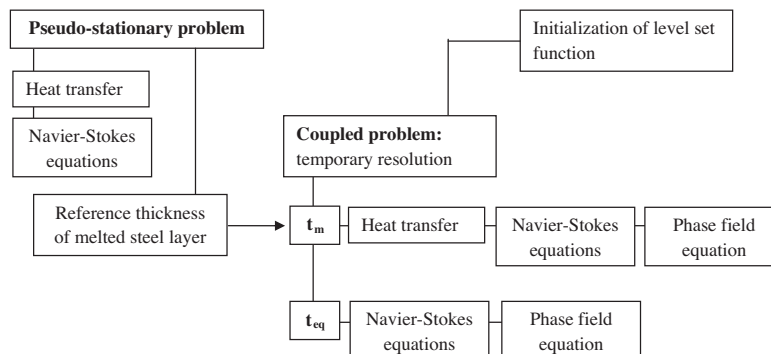


Fig. 2. Resolution algorithm.

the model can be easily adjusted by the variation of Heaviside function parameters dT and dV_m and of interface thickness ε . Good convergence and stability provided by COMSOL Multiphysics solving tools allows multiphysical resolution with strong coupling between heat transfer, fluid flow and phase field problems.

The limitations of the model consist in the application limited to the horizontal plane, in the neglecting of vertical convection movements (buoyancy force) and of possible interdiffusion between immiscible liquids.

4. Results and discussion

The present model provides a big number of results concerning the welding process (temperature field, heat flow, velocity field, phase field, surface tension, etc.), among which the most important is the evolution of morphology and of melted area in time.

The difference in materials properties traduces by the asymmetry of melted zone and by the differentiation of melting process at the each side: while the quantity of melted steel rises constantly in time, copper undergoes partial solidification and remelting. This effect is the most pronounced in case of centered beam position (Fig. 3). In the beginning of thermal cycle, copper dominates in melted zone (A) as it melts more quickly than steel ($t < 0.01$ s). Then, due to high thermal conductivity of solid copper surrounding the melted zone, melted copper area reduces ($0.01 < t < 0.03$ s). Due to heat transfer from steel-rich flow, the advancement of mixing leads to partial copper remelting (B) ($0.03 < t < 0.042$ s). After that, copper-rich liquid area reduces again ($0.048 < t < t_m$), while steel-rich area rises till the end of calculation (C). Consequently, the thickness of melted zone is determined by the propagation of liquid copper area in the beginning and of liquid steel area at the end of thermal cycle.

According to the model, the mixing process in melted zone is controlled by the competitive recirculation flows having relatively low Reynolds numbers ($Re < 10$) and is determined by the position of heat source relatively to joint line and by welding speed. The influence of these parameters on morphology and on the periodical character of solidification is studied.

The offset of heat source from the joint line is a key-parameter controlling the heterogeneity of the liquid zone. The offset of the keyhole to steel side is considered as negative ($-d$) and to copper side as positive ($+d$). In function of this offset, several types of morphology can be observed (Fig. 4):

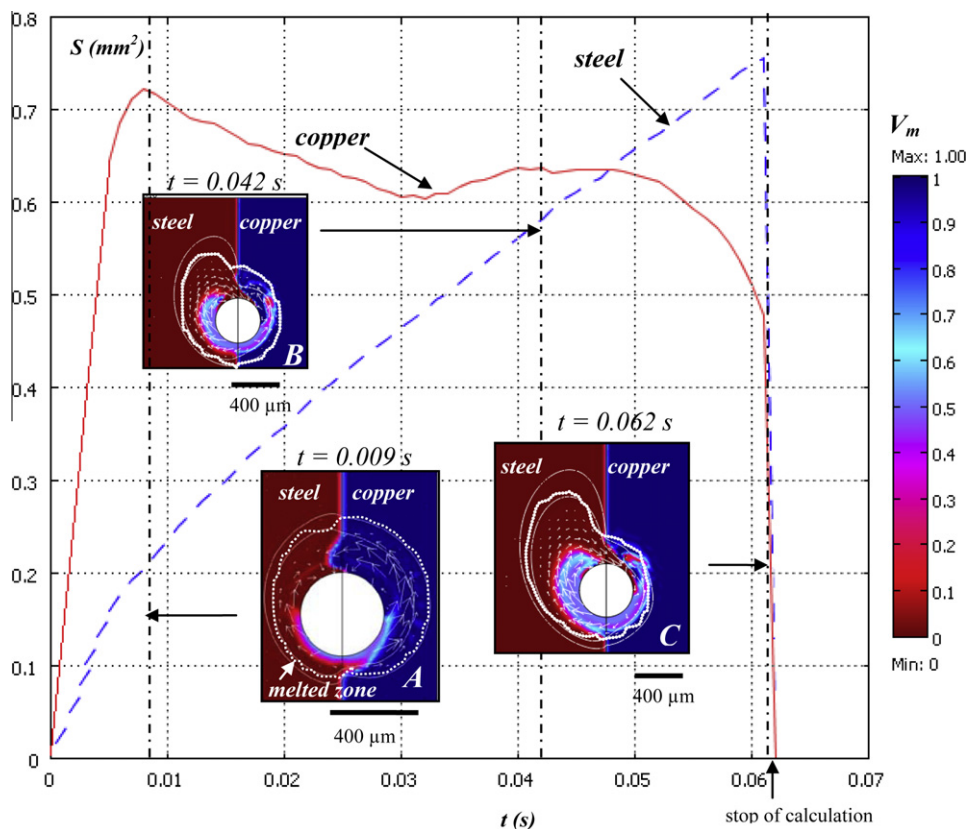


Fig. 3. Evolution of melted area corresponding to each material ($d = 0 \mu\text{m}$, $v_w = 0.9 \text{ m/min}$).

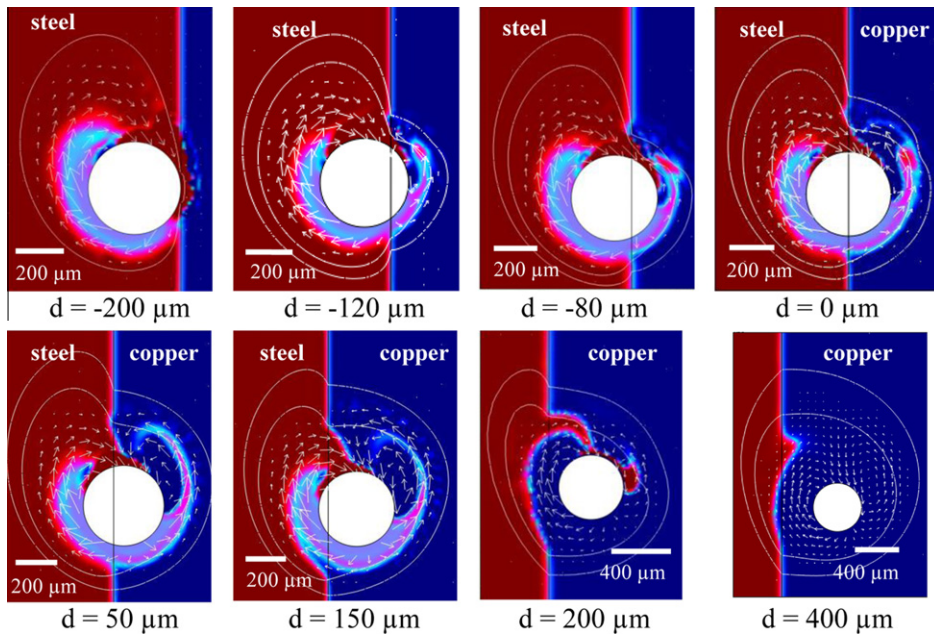


Fig. 4. Evolution of weld morphology in function of heat source position, $v_w = 0.9$ m/min.

- $d = [-200; -100]$ μm : An important recirculation flow rich in steel makes a clockwise movement around the keyhole and enters in copper domain. Copper penetrates in melted zone before the keyhole and forms the clockwise moving zone of intense mixing with steel. However, the global content of copper in melted zone does not exceed 8 at.%. Bigger negative offset makes joining impossible.
- $d = [-120; -80]$ μm : A steel-rich clockwise flow concurrences with less advanced flow of copper; the zone of mixing becomes rich in copper and almost symmetrical. Steel inclusions migrate at copper side and show the tendency to solidify before the rest of melted zone, as they approach solid copper having higher thermal conductivity. Under the action of local velocity field, they take form of thin bands, which are well distinguishable in copper medium.
- $d = [0; -150]$ μm : With a rise of offset values, a counter-clockwise flow rich in copper and the mixing zones become more and more important. Consequently, steel-rich bands may even collide with a main steel-rich flux.
- $d = [200; 500]$ μm : Two concurrent copper-rich flows can be observed. The melted steel layer is pushed out far past the keyhole by one of these flows, where it undergoes the fragmentation when enters in more or less quick recirculation flow. Maximal copper amount when joining is still possible is about 95 at.%.

The morphological elements issued from the model are in good correspondence with experimentally observed mixing patterns of copper–steel joints ([17] and Fig. 7).

The variation of heat source velocity does not create new morphological elements in calculated melted zone. A rise of v_w lengthens the isotherms of fusion and intensifies the convection, which traduces by the development of more significant mixing zones before the keyhole and of advanced immiscible flows in the tail of the melted zone (Fig. 5).

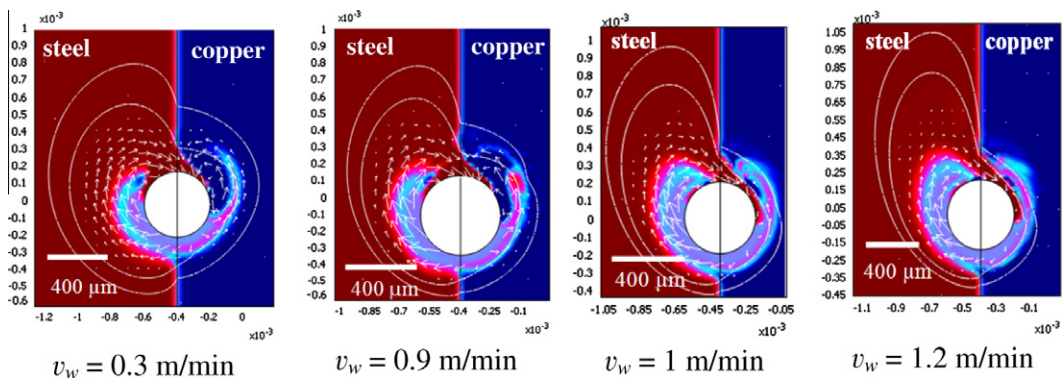


Fig. 5. Evolution of weld morphology in function of heat source speed, $d = 0$ μm .

The composition of melted zone influences the step of heterogeneous pattern. Under fixed velocity of welding ($v_w = 0.9$ m/min), the pattern step diminishes from maximal value (1005 μm) for copper-rich weld ($d = 400$ μm) to minimal (540 μm) for steel-rich weld ($d = -200$ μm) (Fig. 6(a)). The length of melted zone shows the same tendency (Table 2), which maintains an overlap between melted zones in the range of 37–51%.

The study of pattern step in function of welding speed shows close values for low welding speed and sharp rise when v_w comes near to 1 m/min (Fig. 6(b)). Under $v_w > 1$ m/min the step of heterogeneous pattern becomes close to the length of melted zone (Table 2), which signifies the transition from periodical to pseudo-stationary development of melted zone [8]. It can be concluded that v_b is close to 1 m/min.

The information on length and morphology of melted zone issued from the model allows recreating entire repetitive pattern by the superposition of liquid/solid and copper/steel interfaces with a step ΔL . The results of such reconstruction for centered beam position were validated on two examples of copper–steel welds carried out by Nd:YAG laser and electron

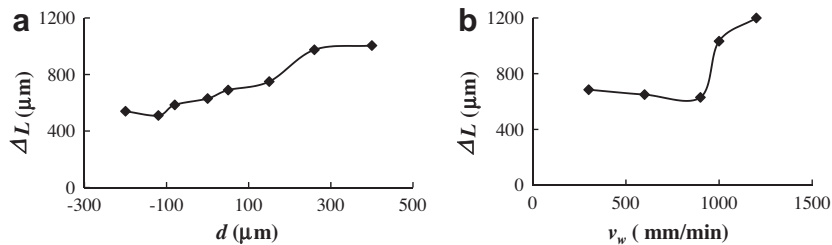


Fig. 6. Variation of pattern step in function of heat source position (a) and welding speed (b).

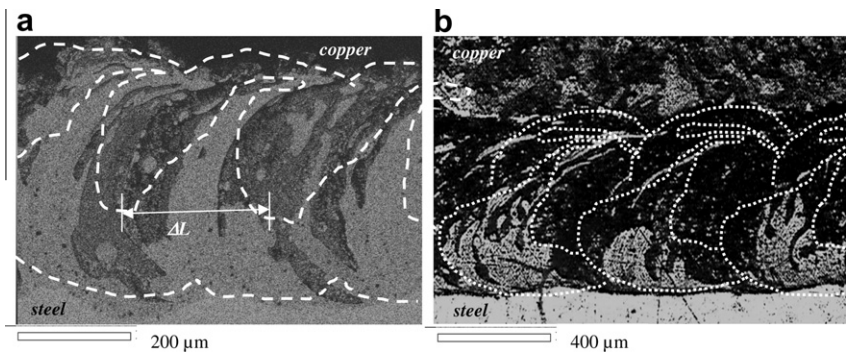


Fig. 7. Comparison of horizontal cuts of copper-AISI 316L stainless steel welds (SEM) with pattern images issued from calculations (solid lines represent calculated position of copper/steel interfaces, when dashed lines show the limits of melted zone): (a) laser weld: $d = 0$ μm , $v_w = 0.5$ m/min, $\varnothing = 200$ μm ; (b) electron beam weld: $d = 0$ μm , $v_w = 0.9$ m/min, $\varnothing = 400$ μm .

Table 2
Operational parameters and some characteristics of welding process.^a

d (μm)	v_w (mm/min)	t_m (s)	ΔL (μm)	L (μm)
-200	900	0.036	540	1100
-120	900	0.034	510	1000
-80	900	0.039	585	1000
0	900	0.042	630	1000
50	900	0.046	690	1100
150	900	0.050	750	1200
260	900	0.065	975	1400
400	900	0.067	1005	1600
0	300	0.137	685	1050
0	600	0.065	650	1000
0	900	0.042	630	1000
0	1000	0.062	1033	1200
0	1200	0.060	1200	1200

^a d - keyhole offset; v_w - welding speed; t_m - melting time; ΔL - pattern step; L - length of melted zone.

beam. Four criterions were chosen for the comparison: the global composition of melted zone, the thickness of the weld, the pattern step and the position of morphological elements in heterogeneous pattern.

Comparing to the model, the real metal flow always has local instabilities, inclusions of the opposite material and some variations of pattern step. In Nd:YAG laser welding (Fig. 7(a)), horizontal pattern is formed by periodical overlap of copper-rich and steel-rich flows. The solid lines presenting copper–steel interfaces and the dashed lines corresponding to the limits of melted zone show a good correspondence in pattern step (210 and 250 μm in real and calculated patterns, respectively) and in morphology. However, in the real weld the intersection of the fluxes is more pronounced: more intensive mixing leads to the propagation of copper flow till solid steel side. Calculated thickness of melted zone has 19% error comparing to the real weld (570 and 650 μm in real and calculated melted zones respectively), which can be explained by insufficient data on the variation of physical properties with temperature. However, the proportion between melted materials is well respected by the model: 49 at.% in real weld (from EDS square analysis) and 46 at.% according to calculation.

In electron beam weld (Fig. 7(b)), a higher speed of heat source results in more intense recirculation movements and in the formation of isolated steel-rich bands near solid copper interface, well reproduced by the model. With higher speed of welding, the pattern step is larger (540 and 630 μm in real and calculated patterns respectively). Bigger diameter of heat source has no influence on the precision in calculation of weld thickness (15% error, 720 and 870 μm in real and calculated patterns respectively). The calculated global composition (54 at.%) is in good correspondence with experiment (58 at.%).

5. Conclusions

A numerical multiphysical model reproducing fusion and mixing process between the dissimilar materials, which are immiscible in liquid state and have a sharp difference in thermophysical properties, has been proposed. By simultaneous calculation of temperature, fluid flow and phase field equations, it allows following the development of melted zone morphology.

The influence of operational parameters on the morphology of melted zone has been studied. The mixing process is controlled by keyhole offset from joint line. A character of repetitive heterogeneous pattern formed in high power beam copper–steel joints is determined from one side by altered fusion of each of those materials in a single melted zone and from other side by pattern step length, which strongly depends on keyhole offset and on linear velocity of heat source. The combination of pseudo-stationary and temporary formulations of heat transfer has allowed estimating the pattern step and critical welding speed corresponding to transition between periodic and stationary solidification modes. The numerical results have been shown to be in good correspondence with SEM observations and EDX analysis of real copper–steel joints performed by laser and electron beam welding.

More realistic representation of welding process requires detailed information on keyhole shape and the introduction of interdiffusion across the moving interface (with respect to limited solubility in studied system). Modeling in 3D may give additional information on vertical convection determined by the difference in materials density, but it will require important computational resources.

One of the perspectives of this study may be the comparison of our calculation results obtained with phase field method by COMSOL Multiphysics with those obtained from volume of fluid method by Fluent.

The model as it is may be applied for studying of mixing between other dissimilar couples with limited solubility: immiscible metallic systems, polymers, liquids etc.

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